

# Unit 8 Pre-Test

## Transformations on the Coordinate Plane

Standards assessed: NC.8.G.2–G.4

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Period: \_\_\_\_\_

SCORE **3.0** Basic Understanding

**NC.8.G.2** Identify transformations and translate figures.  
• *I can name translations, reflections, and rotations and translate a point.*

Match each vocabulary term to its definition, then find each image point.

|                                      |                                                                       |
|--------------------------------------|-----------------------------------------------------------------------|
| <div><div>– ▼</div>translation</div> | <b>A.</b> A flip of a figure over a line.                             |
| <div><div>– ▼</div>reflection</div>  | <b>B.</b> A slide; every point moves the same distance and direction. |
| <div><div>– ▼</div>rotation</div>    | <b>C.</b> A turn of a figure about a fixed point.                     |
| <div><div>– ▼</div>dilation</div>    | <b>D.</b> An enlargement or reduction by a scale factor.              |
| <div><div>– ▼</div>congruent</div>   | <b>E.</b> Same size and same shape.                                   |
| <div><div>– ▼</div>image</div>       | <b>F.</b> The resulting figure after a transformation.                |

Coordinate rules — reference

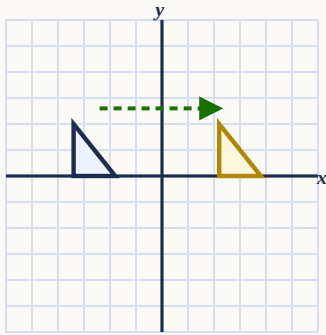
Reflect over x-axis:  $(x, y) \rightarrow (x, -y)$  · Reflect over y-axis:  $(x, y) \rightarrow (-x, y)$   
Rotate 180°:  $(x, y) \rightarrow (-x, -y)$  · Rotate 90° CCW:  $(x, y) \rightarrow (-y, x)$  · Rotate 90° CW:  $(x, y) \rightarrow (y, -x)$   
Dilation (center origin), scale k:  $(x, y) \rightarrow (kx, ky)$

1. Name it

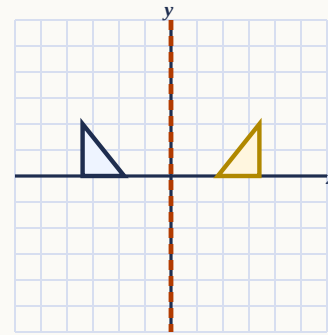
Which transformation is a slide?

2. Name it

Which transformation is a flip?



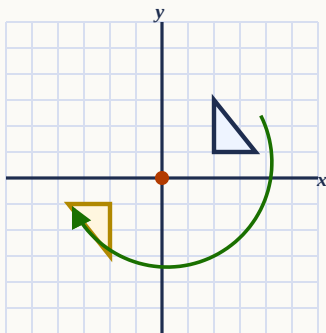
Answer: \_\_\_\_\_



Answer: \_\_\_\_\_

3. Name it

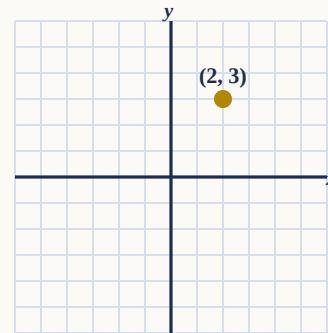
Which transformation is a turn?



Answer: \_\_\_\_\_

4. Translate

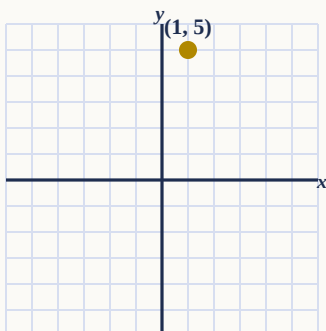
Translate  $(2, 3)$  right 4 units. New point?



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5. Translate

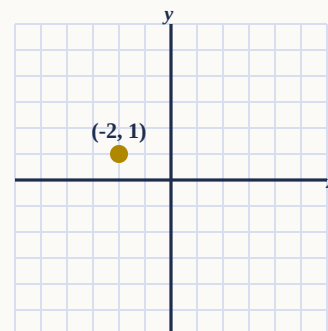
Translate  $(1, 5)$  down 2 units. New point?



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6. Translate

Translate  $(-2, 1)$  left 3 and up 2. New point?



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SCORE **4.0** Target Content

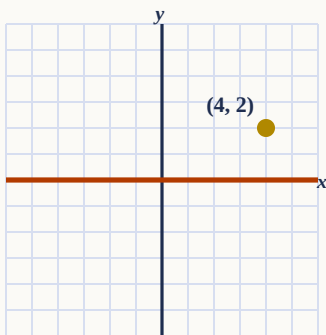
**NC.8.G.3** Reflect and rotate points on the coordinate plane.

- I can find the image of a point after a reflection or rotation.

**Find each image point. Write your answer as (x, y).**

**7.** Reflect

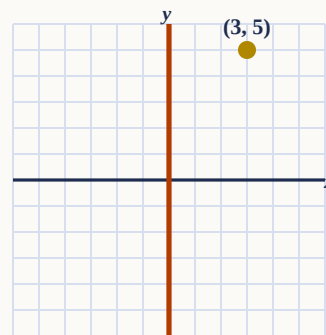
Reflect (4, 2) over the x-axis.



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**8.** Reflect

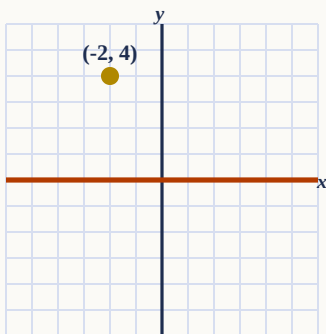
Reflect (3, 5) over the y-axis.



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**9.** Reflect

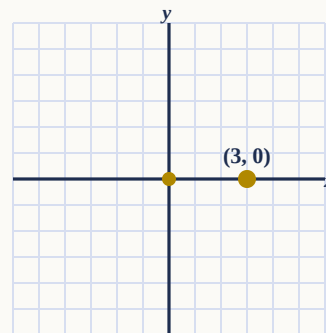
Reflect (-2, 4) over the x-axis.



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**10.** Rotate

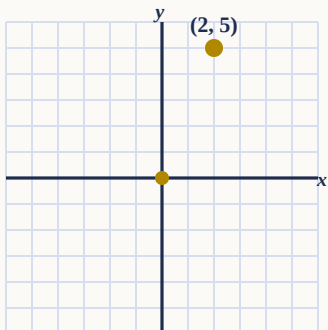
Rotate (3, 0) 90° counterclockwise about the origin.



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11. Rotate

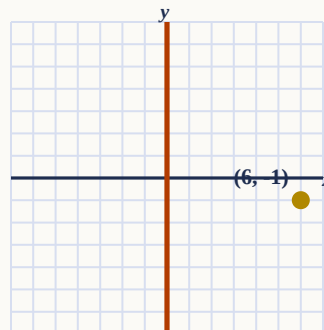
Rotate  $(2, 5)$   $180^\circ$  about the origin.



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12. Reflect

Reflect  $(6, -1)$  over the y-axis.



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SCORE **5.0** Advanced Content

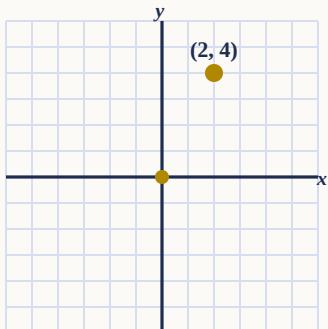
**NC.8.G.4** Dilate figures and describe congruence and similarity.

- I can dilate a point by a scale factor and tell whether figures are congruent or similar.

**Find each image point or answer each question.**

13. Dilate

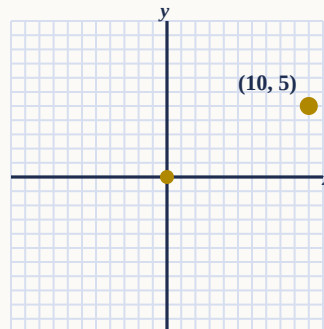
Dilate  $(2, 4)$  by scale factor 3 (center origin).



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14. Dilate

Dilate  $(10, 5)$  by scale factor  $\frac{1}{2}$  (center origin).

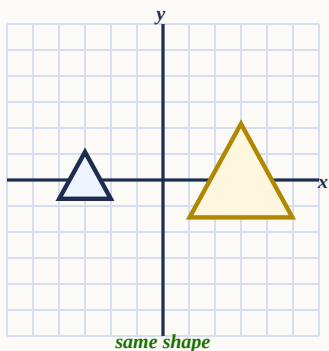


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15. Scale

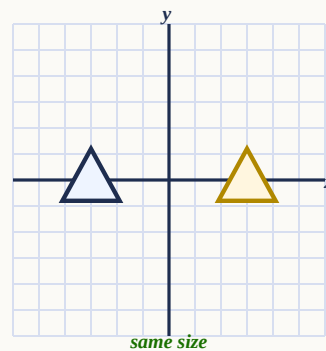
16. Congruence

A dilation with scale factor 2 makes the image \_\_\_\_ (larger or smaller).



Answer: \_\_\_\_\_

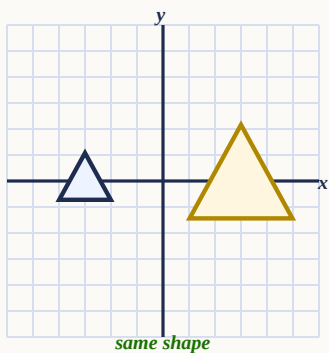
After a translation, the image is \_\_\_\_ to the original (congruent or similar).



Answer: \_\_\_\_\_

**17. Similarity**

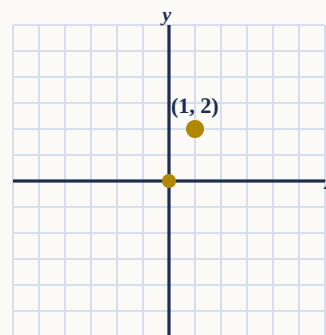
After a dilation with scale factor  $\neq 1$ , the image is \_\_\_\_ to the original.



Answer: \_\_\_\_\_

**18. Rotate**

Rotate  $(1, 2)$   $90^\circ$  clockwise about the origin.



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